

NTSC PAL Video Encoder

Version 1.0



SHMT Co., Ltd.

REVISION HISTORY

Version	Detail(굴림/Verdana 10pt bold)	Name	Date
V.01	Core block specification	NamKwol Kang	Jan .1 st . 2007

Technical documents

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1. NTSC PAL Video encoder

1.1. Overview

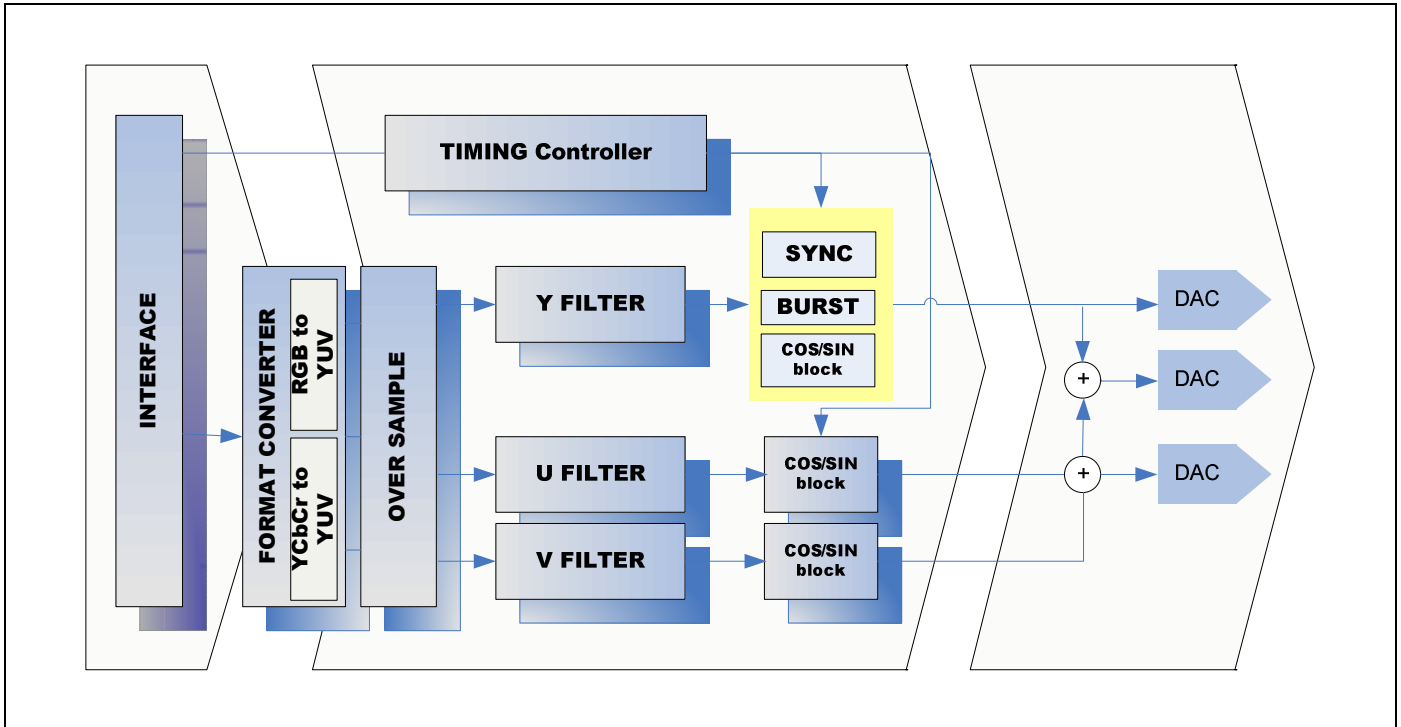
High performance video encoder supports for NTSC M, NTSC-J, NTSC4.32, PAL B/D/G/H/I and PAL M. By using RGB format, it outputs analog composite and analog S-video. This encoder drives the 10bits DAC

1.1.1. Feature

- RGB to PAL/NTSC video encoder
- NTSC M, NTSC J, NTSC 4.43, PAL Nc, PAL B/D/G/H/I, PAL-M
- 27Mhz clock required (x2 oversampling)
- Video input data port supports:
 - RGB 24bit Parallel input format
- Video output supports:
 - Composite / S-Video signal output
 - 10bits high quality DAC output
- Programmable luma / chroma filters
- Programmable subcarrier frequency and phase
- Programmable LUMA delay
- Individual on/off control of each DAC
- CCIR and square pixel operation
- Color signal control/burst signal control
- Complete on-chip video timing generator
- On-chip color bar / color lamp generation

1.1.2. Block diagram

Figure 1-1 Video Encoder Block Diagram



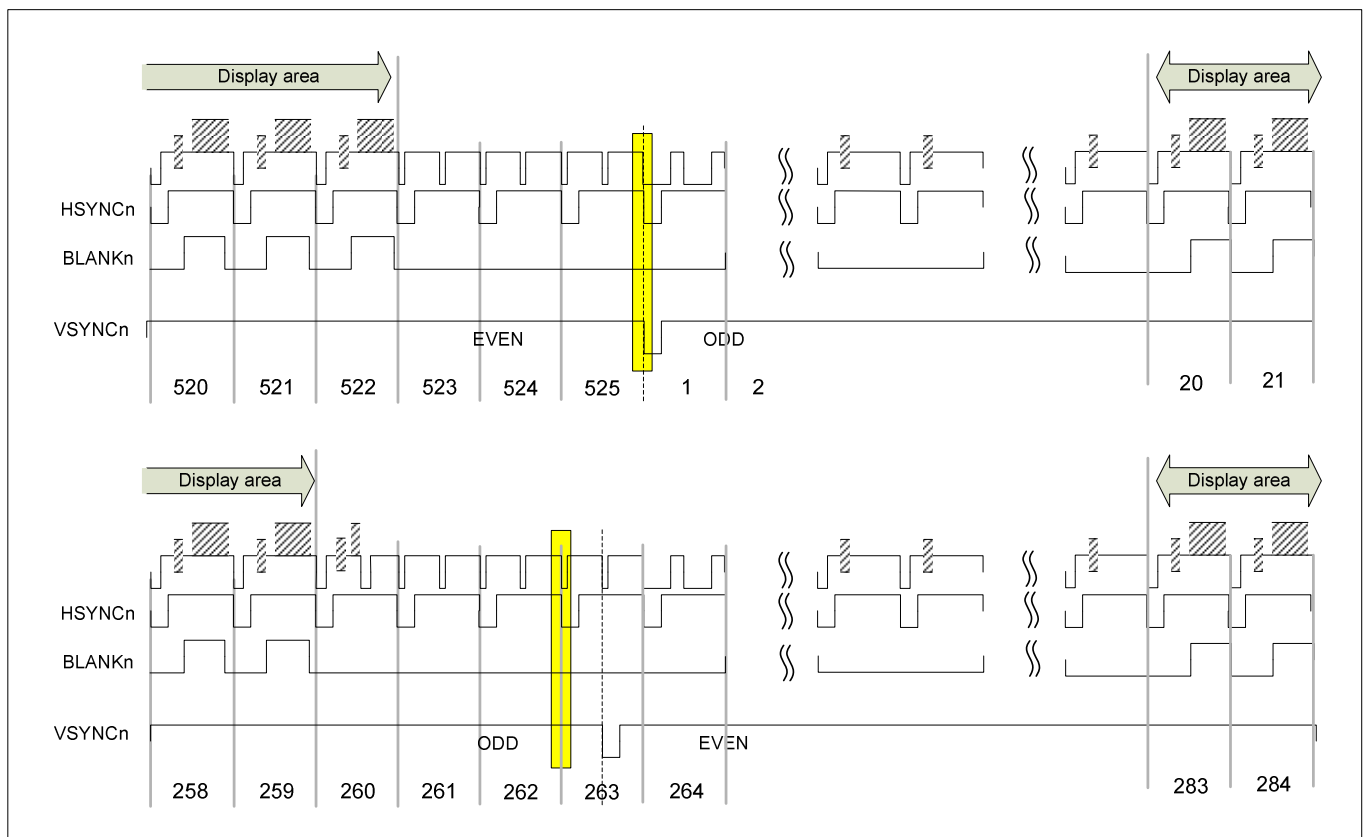
1.2. Operation

1.2.1. Video timing

1.2.1.1. Data input timing

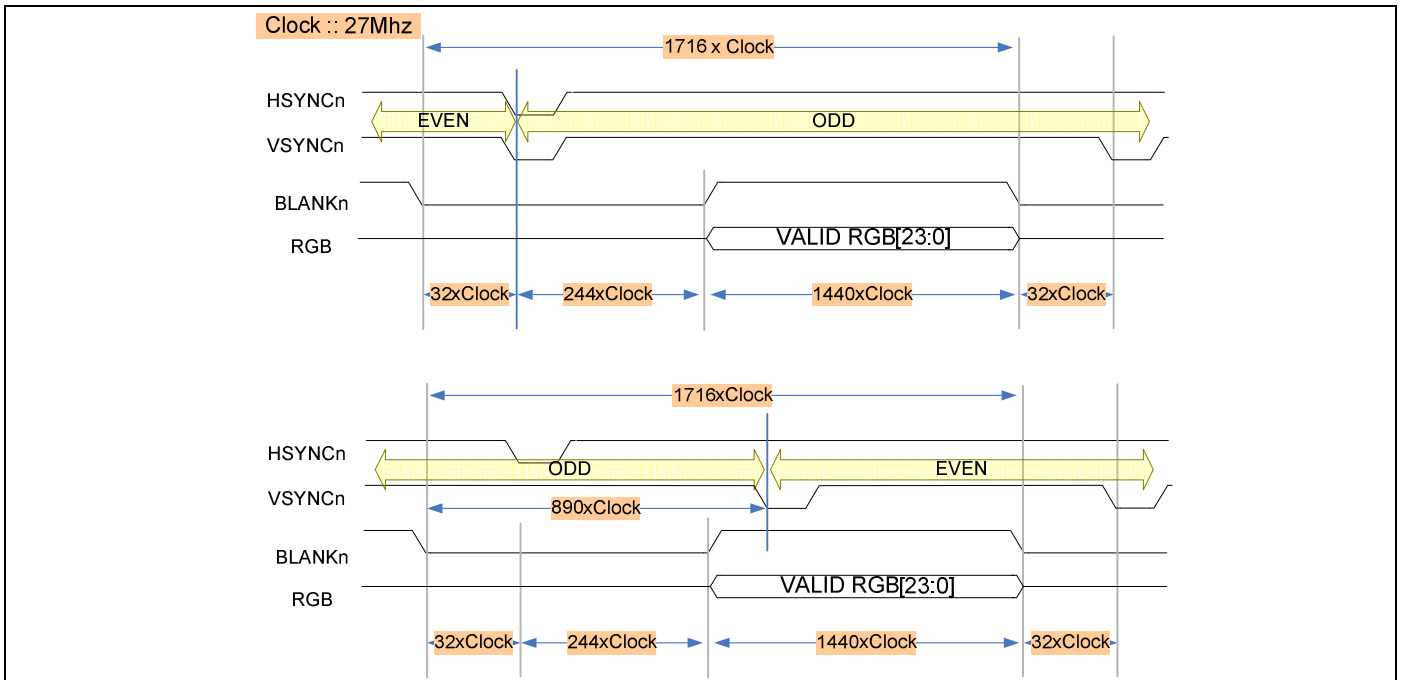
Video input 은 24bits RGB Slave를 지원하도록 되어 있으며, BLANKn = "High" 일 경우, 유효한 정보를 입력 받습니다.

Figure 1-2 Interlaced NTSC Video input timing chart



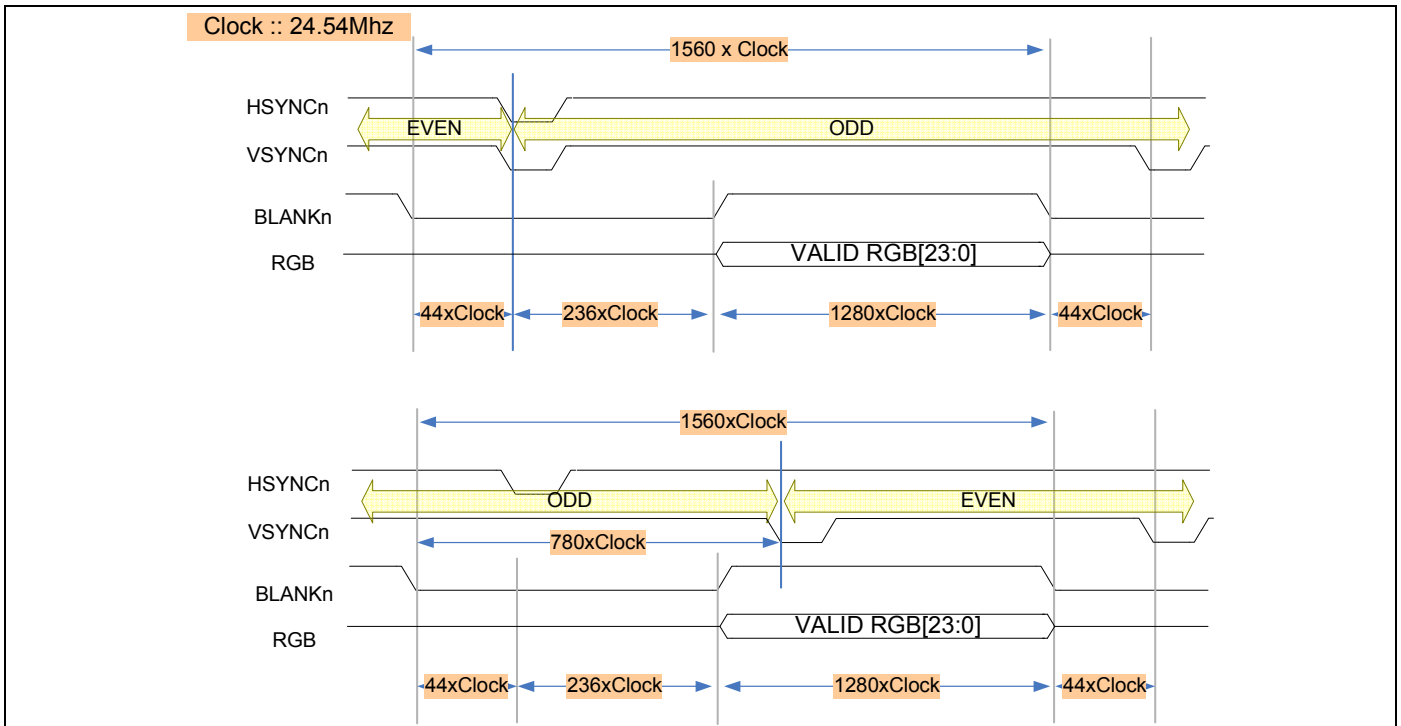
NTSC format 일 경우 Data Active 영역 은 odd field 20~259 / even field 283~522 입니다. Data Active 영역에서 720x480 format에 RGB 신호를 받습니다. Odd와 Even으로 입력 구분은 HSYNCn = "LOW" 와 VSYNC = "LOW" 인 경우를 기준으로 Even → Odd로 변경 됩니다. Even 에서 Odd로 변경 될 경우는 VSYNC= "LOW"인 경우를 기준으로 합니다. 자세한 Timing 은 [Figure 1-3 참조](#)

Figure 1-3 NTSC Even and Odd timing chart



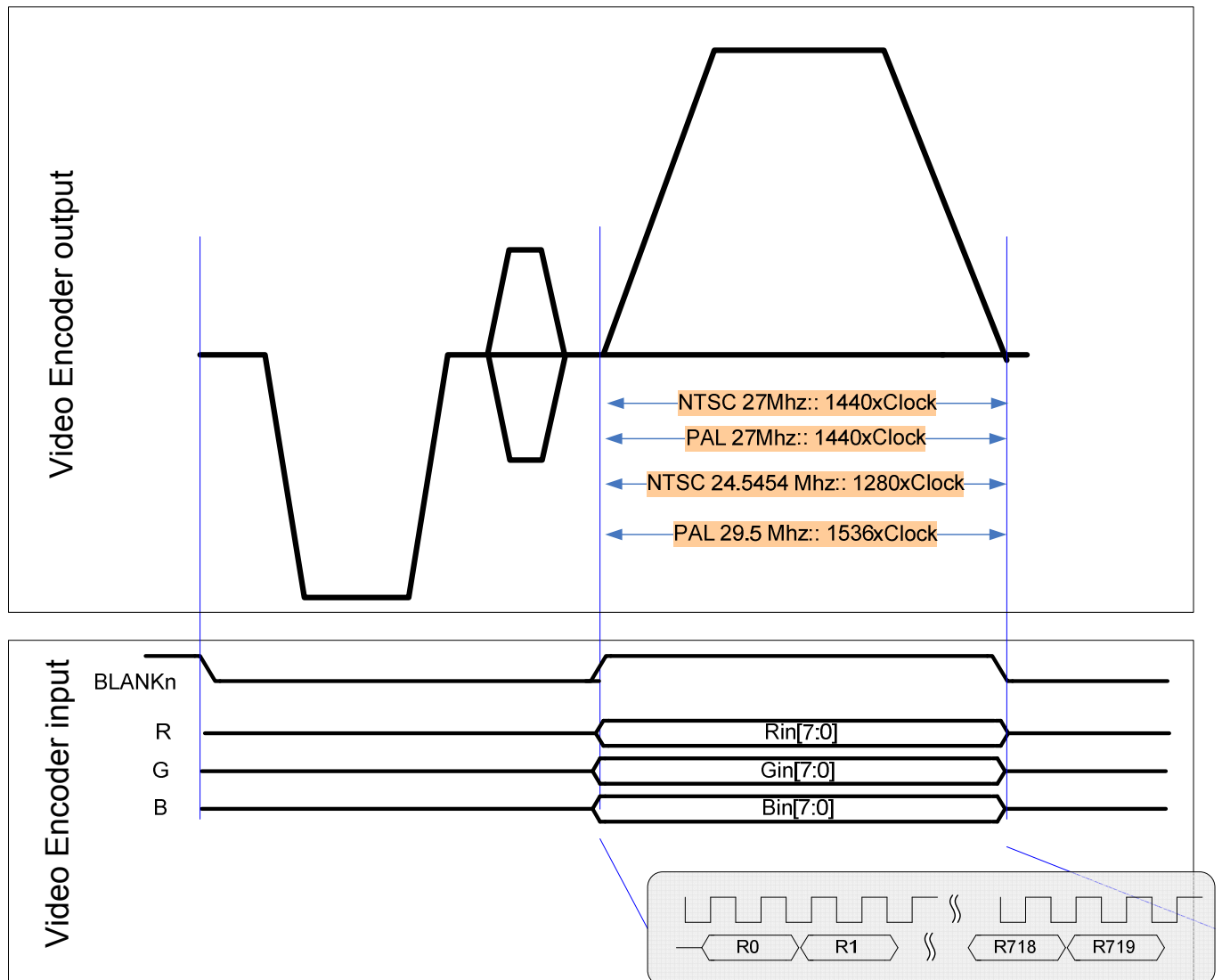
NTSC mode 일 경우 1Line 당 27Mhz sample x 1716 입니다. 출력되는 화상은 720x480 크기를 가지며, Interlaced로 출력 됩니다.

Figure 1-4 Square NTSC Even and Odd timing chart



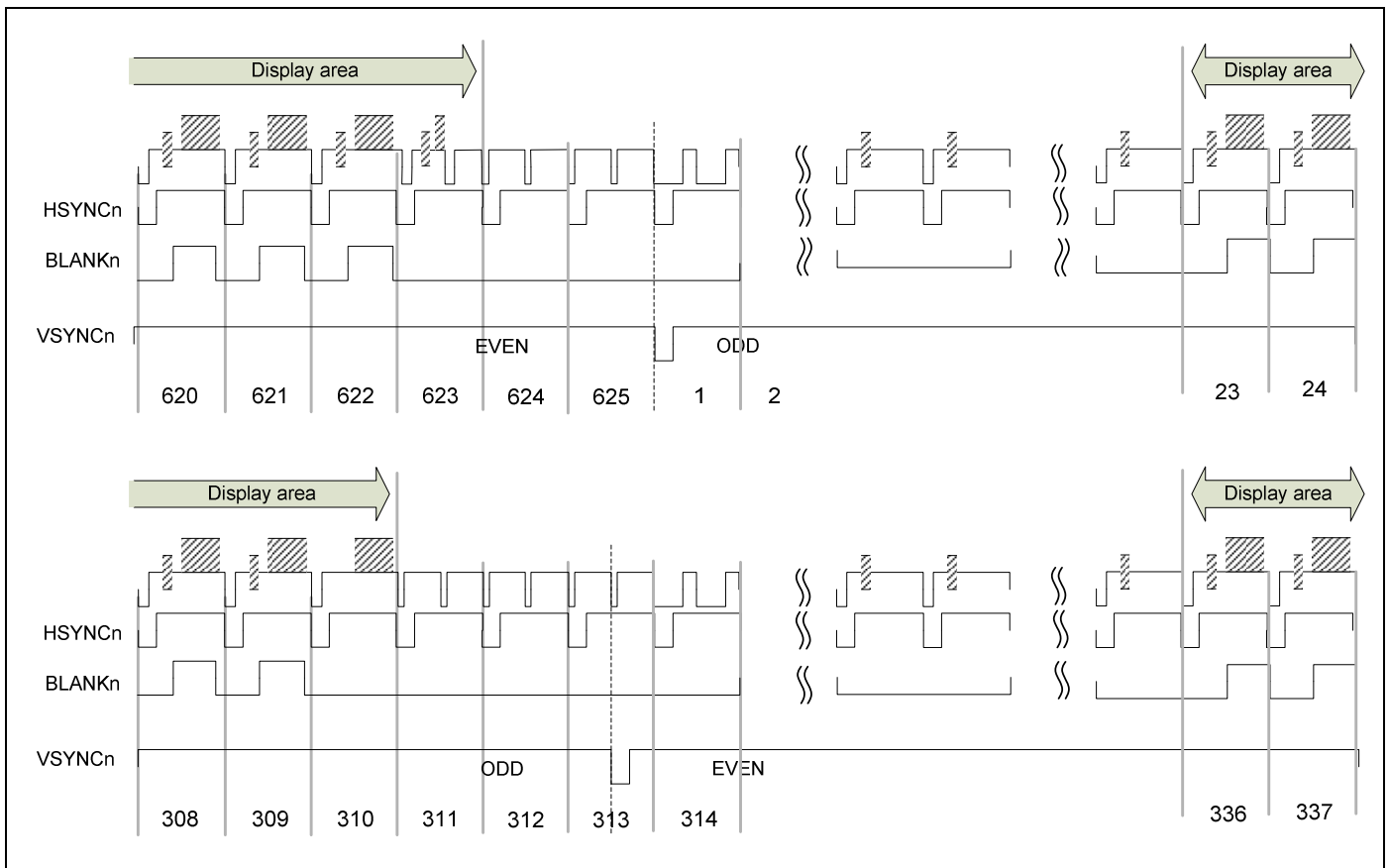
Square mode 일 경우, 24.54Mhz를 공급하며 640x480화상을 출력합니다.

Figure 1-5 RGB input timing



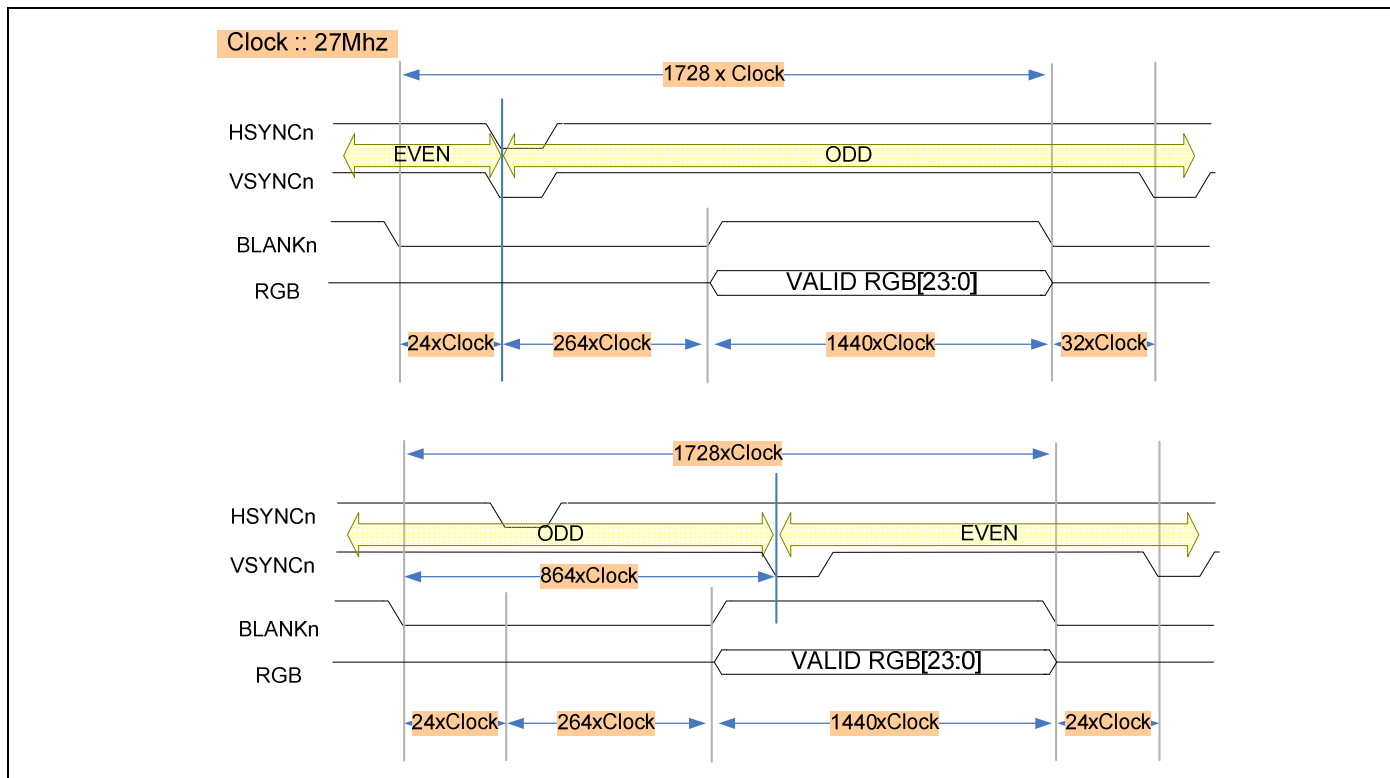
PAL/NTSC 27Mhz 에서 BLANKn 구간은 1440Clock 동안에 총 720 sample를 입력 받도록 되어있으며, Clock에 positive edge 에서 data를 가져 가도록 구성 되어있습니다.

Figure 1-6 PAL Video input timing chart



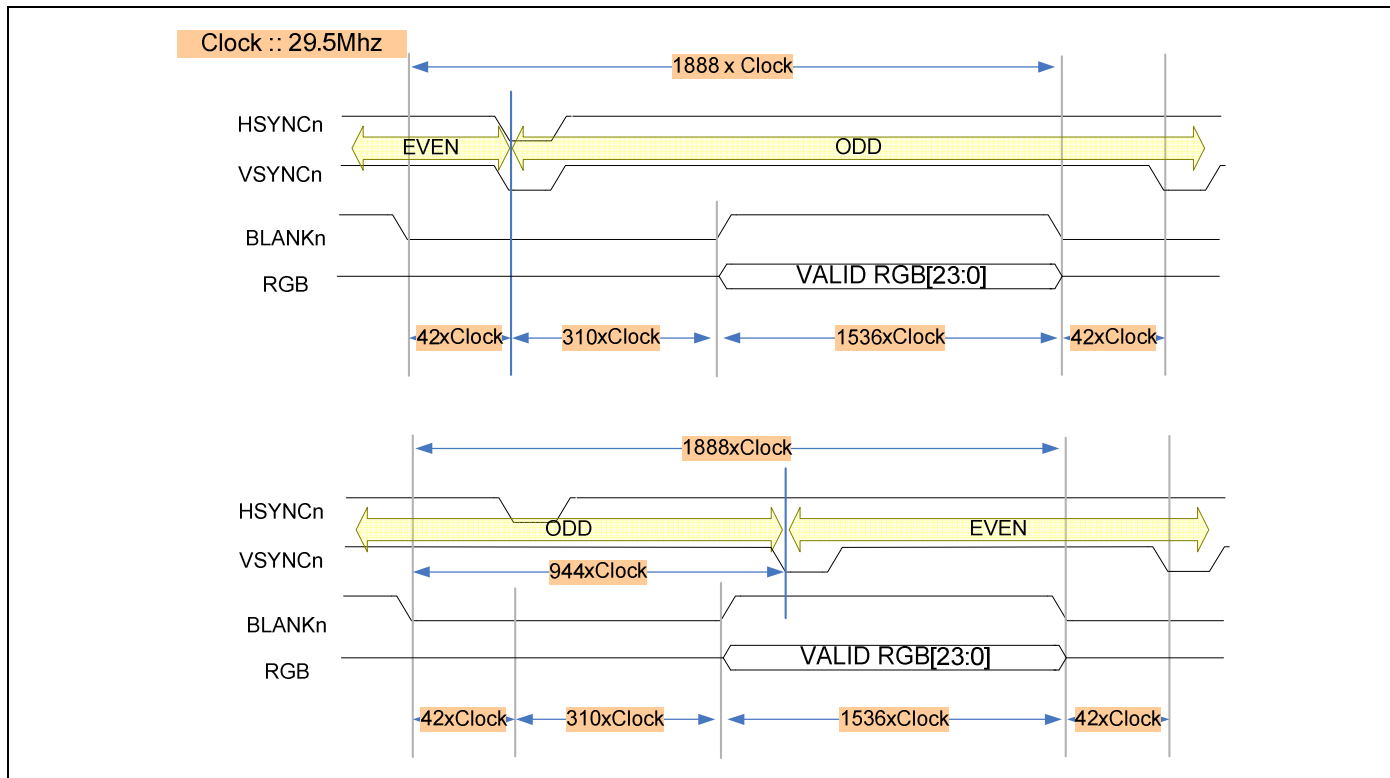
BLANKn = "HIGH" 에서 RGB 입력을 받으며, Data active 영역은 Odd field에서 23~310 총 288 Line Even field에서 336~623 총 288 Line으로 PAL은 720x576 RGB format을 지원합니다.

Figure 1-7 PAL Even and Odd timing chart



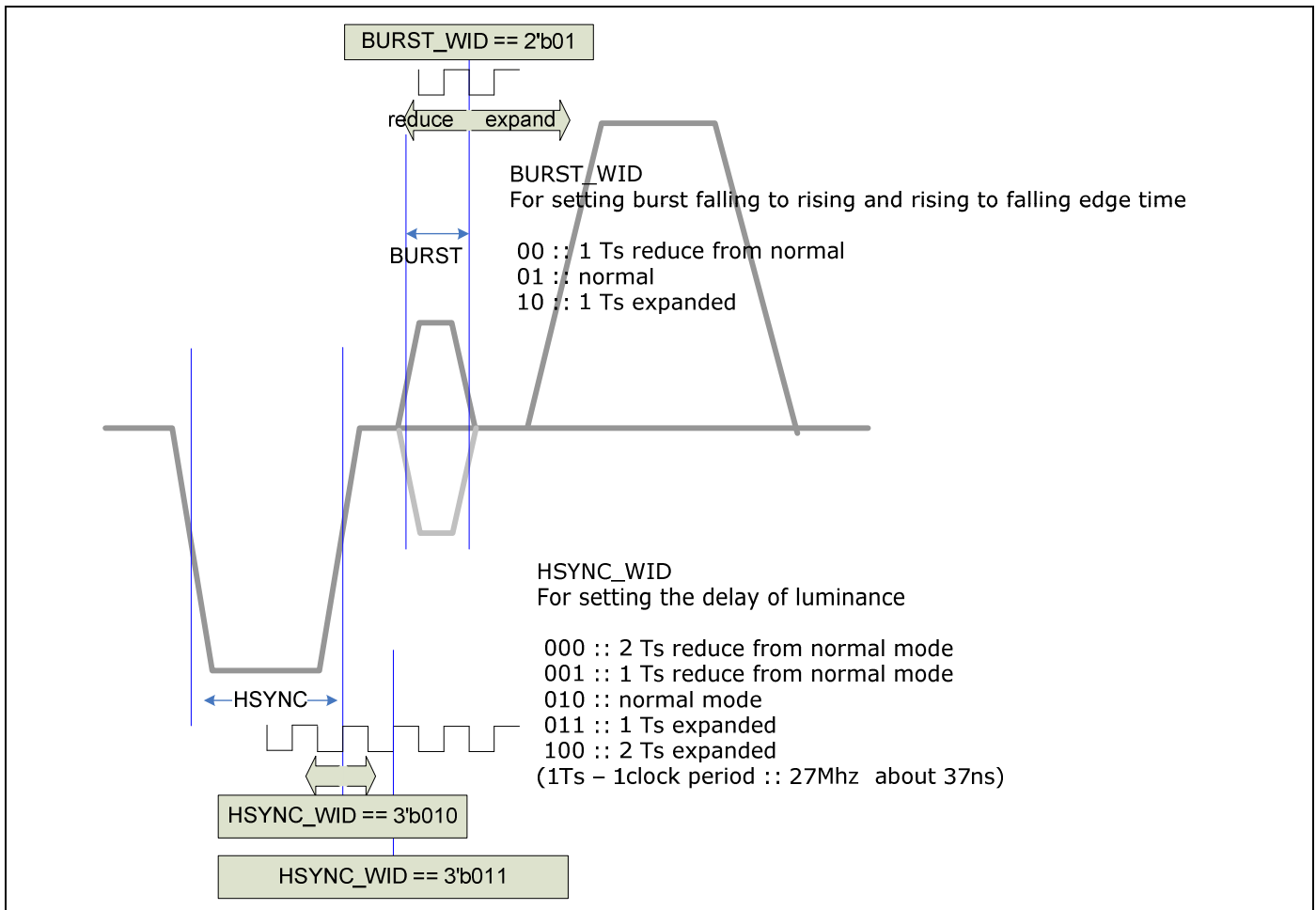
입력 주파수는 27Mhz 를 공급하고, image 576x720 입력합니다. Timing은 Figure 1-6 과 같이 합니다.

Figure 1-8 Square PAL Even and Odd timing chart



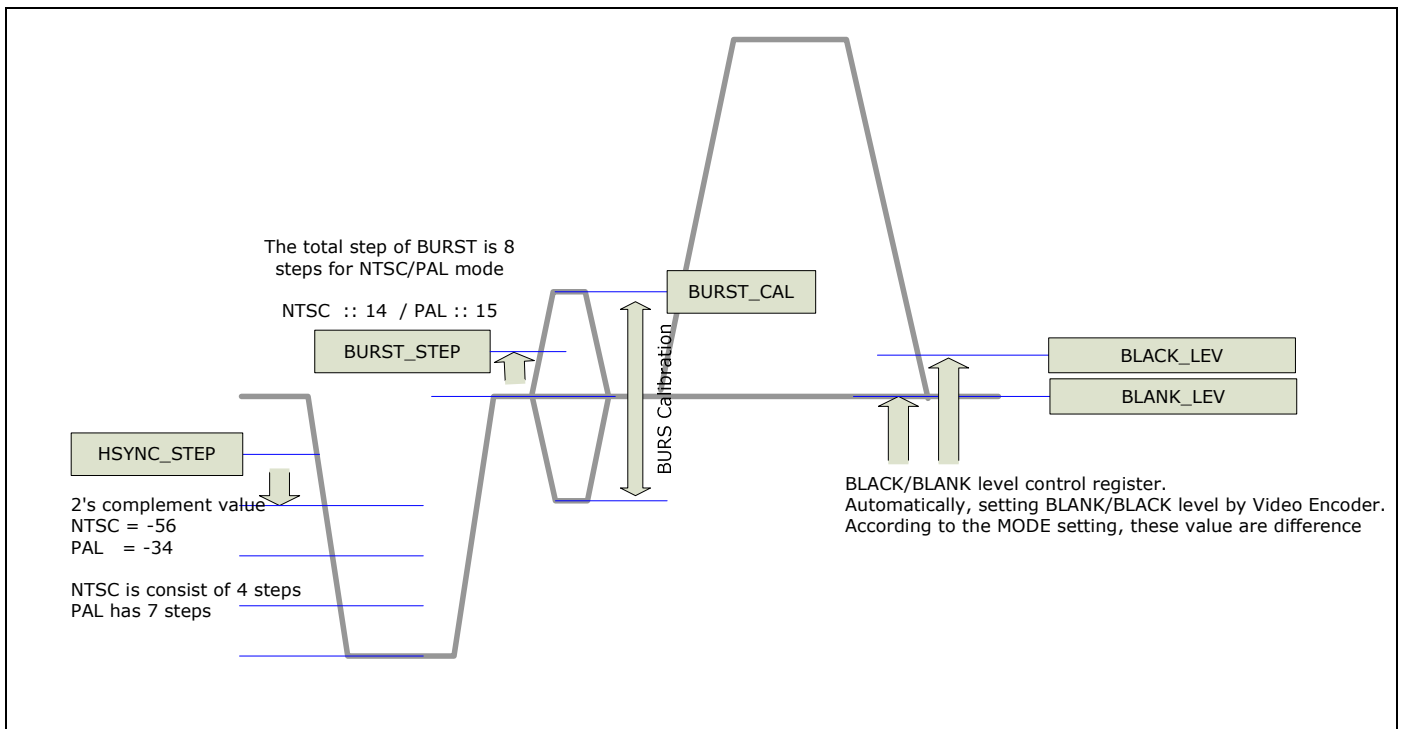
1.2.1.2. Data output timing

Figure 1-9 output HSYNC / BURST timing chart



사용되는 기준 Clock (27Mhz) 를 기준으로 HSYNC width와 BURST width를 조절 할 수 있습니다.
Normal 값보다 1clock 또는 2clock 내외에서 설정 가능 합니다.

Figure 1-10 Output level control chart



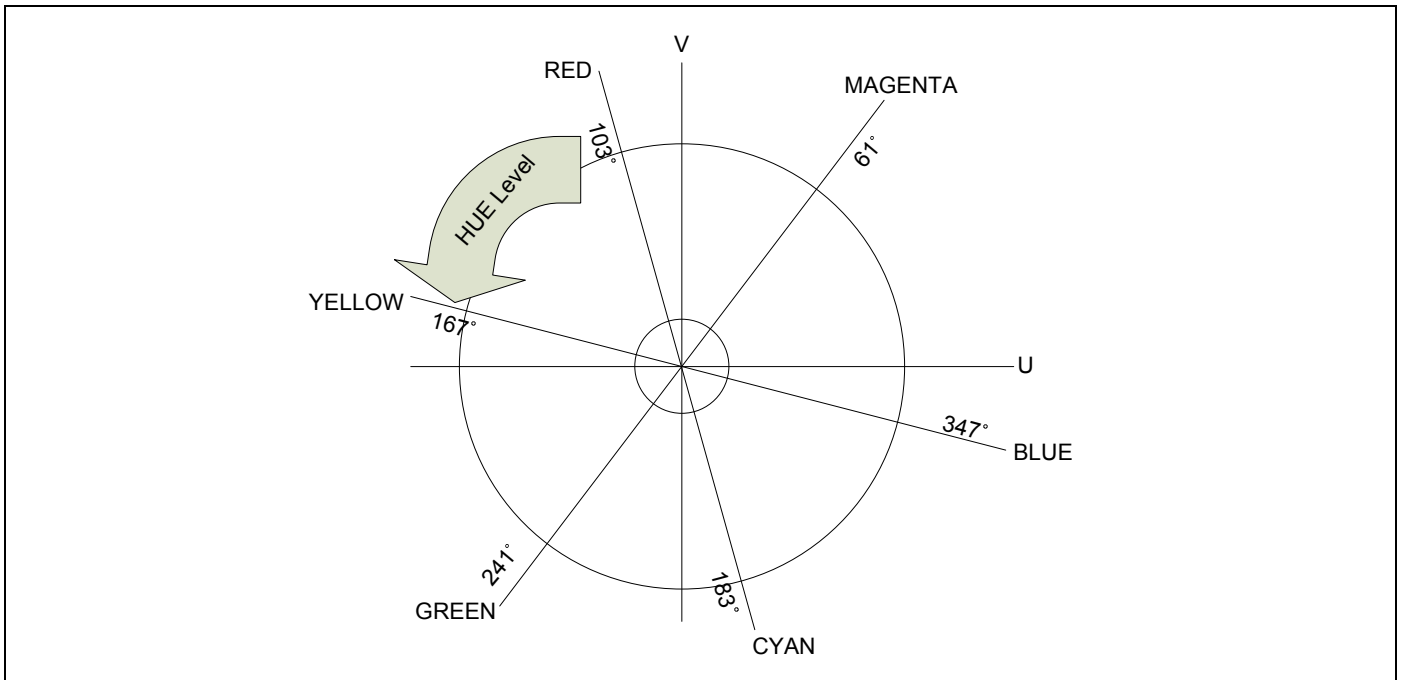
BLACK_LEV 은 NTSC에 경우 7.5IRE가 적용된 값인 282를 적용합니다. PAL 에 경우 252를 사용합니다.

BLANK_LEV 도 NTSC와 PAL경우가 다르며, NTSC → 240 / PAL → 252 가 적용됩니다.

BURST_STEP 은 총 8 step으로 구성되며, 한 Step당 크기를 설정할 수 있도록 되어있습니다.

이와 마찬가지로 HSYNC_STEP 또한 step 당 설정 가능합니다.

Figure 1-11 Hue level control chart



RED 신호를 YELLOW 출력으로 변경할 경우, HUE_LEV 을 이용하여 변경 가능합니다. RED와 YELLOW에 차는 $167^{\circ} - 103^{\circ} = 64^{\circ}$ 위상 차가 발생합니다. HUE_LEV이 255일 경우 360° 이므로,

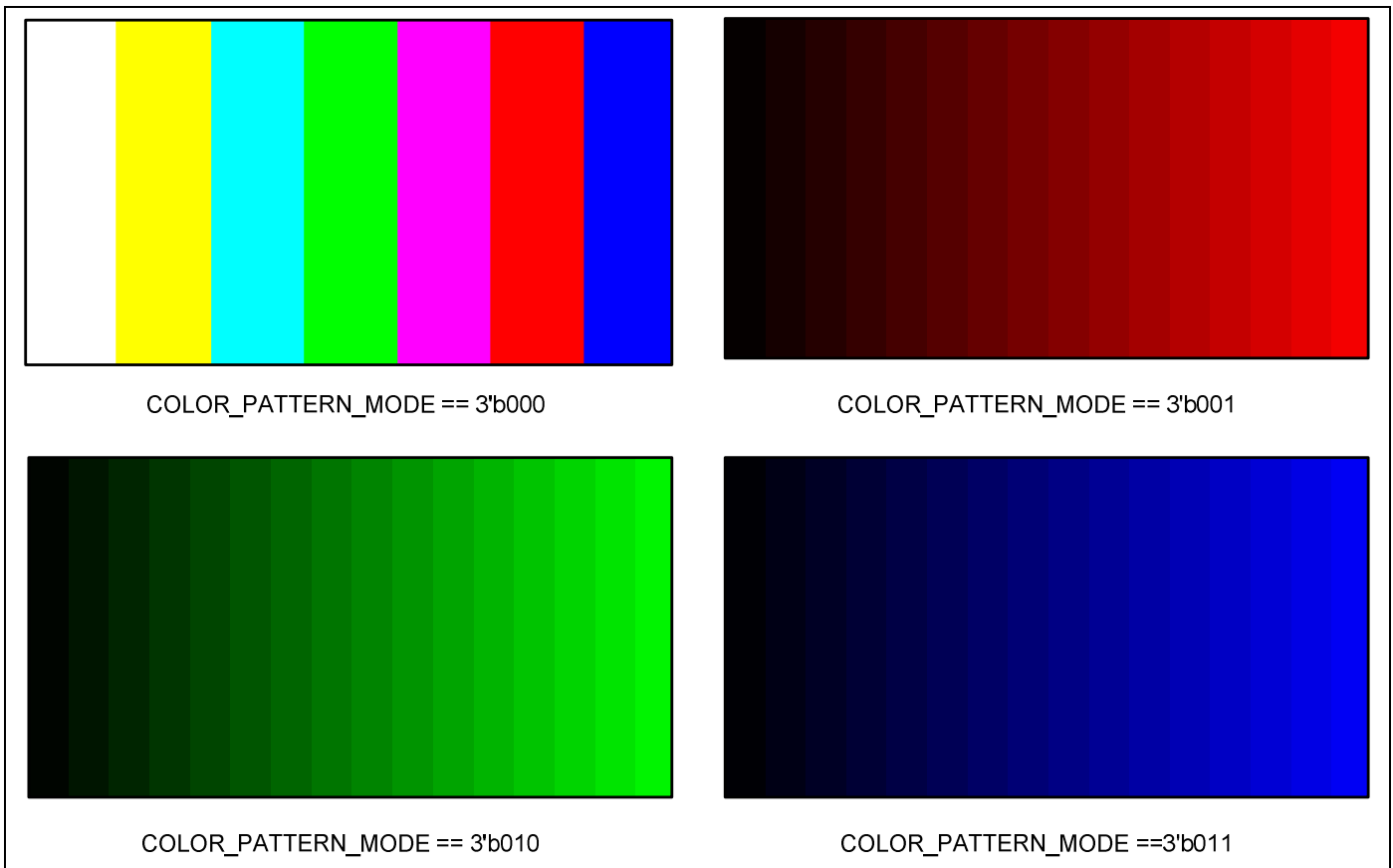
$(64 / 360) \times 255 = 45$ 입니다. 따라서 HUE_LEV = 45를 설정하면, RED Level에 신호들은 Yellow값을 갖습니다. Hue 정보만 수정되기 때문에 Saturation/Luminance 정보는 본래 값을 갖게 됩니다.

1.2.2. Control setting

Video Encoder 동작을 위하여 각 설정 정보를 설정한 후 Enable을 설정 하는 순서로 이루어지며, NTSC Video Encoder는 720x480 image를 기본 영역으로 하는 image를 NTSC(M)을 기본으로 제공하는 TV및 영상 장치에 출력 할 수 있도록 되어 있습니다.

1.2.2.1. Internal Pattern generation

Figure 1-12 Internal pattern



`COLOR_PATTERN_MODE`를 선택 하고, `EN_INTERNAL_PATTERN== 1'b1`로 설정 하게 되면 외부에 입력신호 없이 자체 SYNC 신호 및 RGB 신호를 생성하여, `MODE`에 따라 출력 하게 됩니다.

1.2.2.2. Subcarrier frequency setting

Table 1-1 Subcarrier frequency table

MODE	Name	Setting value (SUB_REQ)	Frequency
3'b000	NTSC M	21F0 78BB h	3.579545 Mhz
3'b001	NTSC J	21F0 78BB h	3.579545 Mhz
3'b010	NTSC 4.43	2A09 8ACB h	4.43361875 Mhz
3'b011	M PAL	21E6 EFA4 h	3.57561149 Mhz
3'b100	PAL(B, D, G, H and I)	2A09 8ACB h	4.43361875 Mhz
3'b101	PAL Nc	21F6 9446 h	3.58205625 Mhz
3'b110	PAL N	2A09 8ACB h	4.43361875 Mhz

SUB_REQ를 각 MODE에 따라서 설정 하여야 합니다. Subcarrier 설정을 위하여 각 값들을 설정 하게 됩니다.

Register 계산방법은 아래와 같습니다.

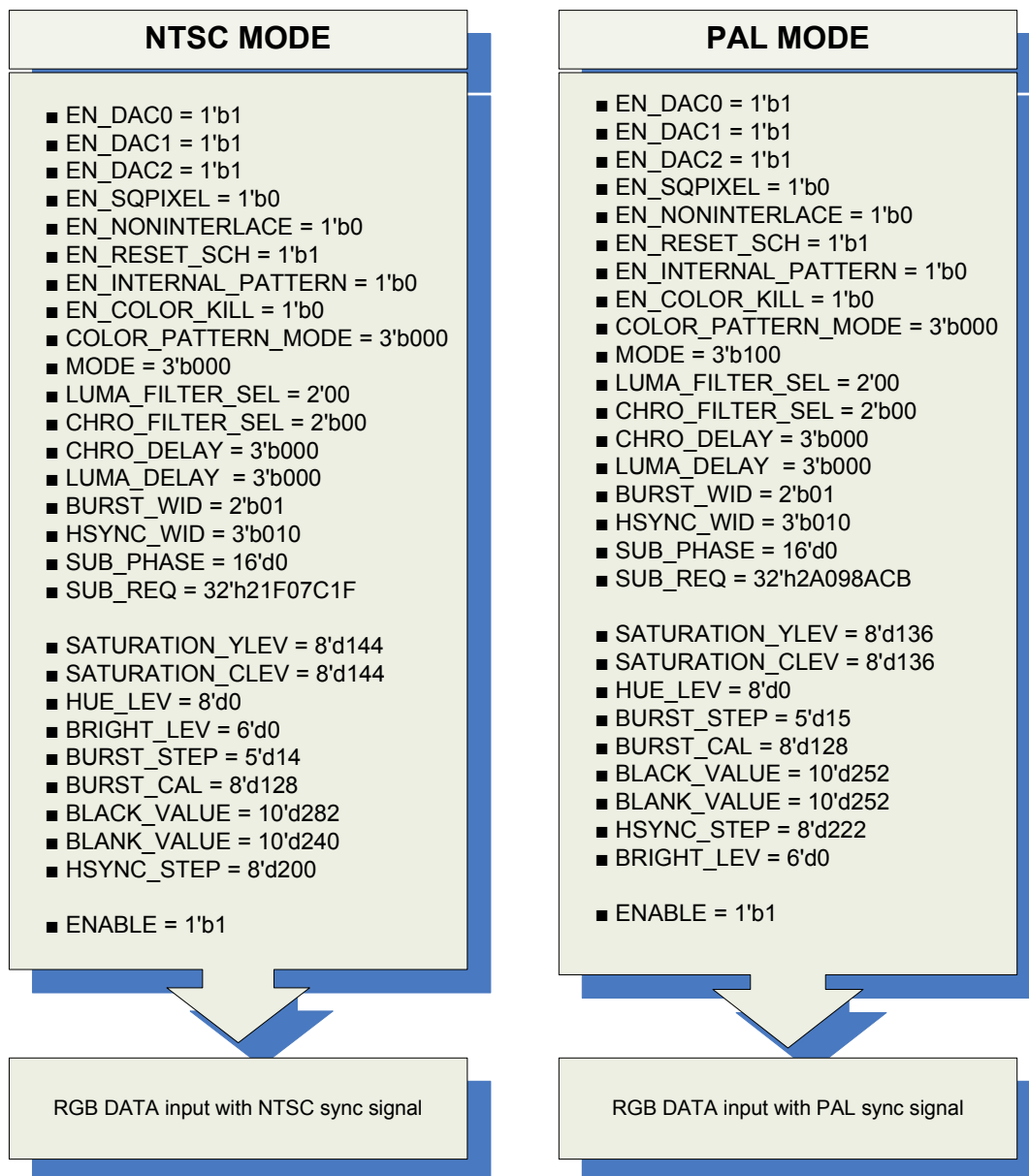
$$\begin{aligned}
 & \left\{ \left(\text{SubCarrier frequency (3.57954Mhz)} \times \text{Total Clock (NTSC :: 1716 / PAL :: 1728) in Hsync} \right) / \text{Input frequency (27Mhz)} \right\} \times 2^{32} \\
 & \quad \text{Total Clock (NTSC :: 1716 / PAL :: 1728) in Hsync} \\
 & = \frac{\text{SubCarrier frequency (3.57954Mhz)}}{\text{Input frequency (27Mhz)}} \times 2^{32}
 \end{aligned}$$

예를 들어, 3.57954Mhz에 Subcarrier를 사용하고, 입력 주파수가 27Mhz 일 경우,

$$\frac{3.57954\text{Mhz}}{27\text{Mhz}} \times 2^{32} = 569407675 = 32'h21F078BB$$

1.2.2.3. Mode setting value

Figure 1-13 NTSC / PAL setting



1.3. Signal Description

1.3.1. InOutput Signal

Table 1-2 VideoEncoder Signal

Signal Name	Direction	Description	Init. value
CLK	input	Input clock (27Mhz)	
RESETn	Input	Reset	
HSYNCn	Input	Horizontal sync signal	
VSYNCn	Input	Vertical sync signal	
BLANKn	Input	Blank sync signal	
Rin	Input	R[7:0] Red color	
Gin	Input	G[7:0] Green color	
Bin	Input	B[7:0] Blue color	
ENABLE	Input	Enable signal	
EN_DAC0	Input	Composite enable signal	
EN_DAC1	Input	Luminance enable signal	
EN_DAC2	Input	Chrominance enable signal	
EN_SQPIXEL	Input	Square pixel enable signal	
EN_NONINTERLACE	Input	Non-interlace mode signal	
EN_INTERNAL_PATTERN	Input	Internal color generation	
EN_COLOR_KILL	Input	Composite color remove signal	
COLOR_PATTERN_MODE	Input	Internal Color pattern selection	
MODE	Input	Output encoder format selection	
LUMA_FILTER_SEL	Input	Luminance filter selection	
CHRO_FILTER_SEL	Input	Chrominance filter selection	
CHRO_DELAY	Input	Chrominance delay	
LUMA_DELAY	Input	Luminance delay	
BURST_WID	Input	Burst width setting	
HSYNC_WID	Input	Hsync width setting	
SUB_PHASE	Input	Subcarrier phase setting	
SUB_REQ	Input	Subcarrier frequency setting	
SATURATION_YLEV	Input	Saturation luminance level setting	
SATURATION_CLEV	Input	Saturation chrominance level setting	
BLANK_VALUE	Input	Blank value setting	
HUE_LEV	Input	Hue level setting	
BRIGHT_LEV	Input	Brightness level setting	

Signal Name	Direction	Description	Init. value
BURST_STEP	Input	Burst step setting	
HSYNC_STEP	Input	Hsync step setting	
BURST_CAL	Input	Burst level setting	
BLACK_VALUE	Input	Black value setting	
DAC0_ENABLE	Output	DAC enable signal	
DAC1_ENABLE	Output	DAC enable signal	
DAC2_ENABLE	Output	DAC enable signal	
DAC0_DATA	Output	Composite signal data	
DAC1_DATA	Output	Luminance signal data	
DAC2_DATA	Output	Chrominance signal data	
F_COUNTER_INTER	Output	Field counter	
H_COUNTER_INTER	Output	Horizontal counter	
V_COUNTER_INTER	Output	Vertical counter	

1.4. Setting Input/Output Description

1.4.1. Status input/output

Table 1-3 Status input/output Description

Bits	Name	I/O	Description	Init. value
[7:0]	FIELD_CNT	output	Field counter	0b
[10:0]	H_CNT	Output	Horizontal counter	0b
[9:0]	V_CNT	Output	Vertical counter	0b
[00]	Enable	Input	Enable status	0b

1.4.2. Control input

Table 1-4 Control input Description

Bits	Name	I/O	Description	Init. value
[00]	EN_DAC0	Input	This bit can be used to power down the DAC0 0 :: Off DAC / 1 :: On DAC	0b
[00]	EN_DAC1	Input	This bit can be used to power down the DAC1 0 :: Off DAC / 1 :: On DAC	0b
[00]	EN_DAC2	Input	This bit can be used to power down the DAC2 0 :: Off DAC / 1 :: On DAC	0b
[00]	EN_SQPIXEL	Input	This bit is used for square pixel mode 0 :: normal mode 1 :: Square pixel mode for the square pixel mode, change the frequency. - NTSC square pixel mode :: 24.5454MHz - PAL square pixel mode :: 29.5MHz	0b
[00]	EN_NONINTERLACE	input	To choice the mode of interlace or non-interlace 0 :: interlace 1 :: non-interlace	0b

Bits	Name	I/O	Description	Init. value
[00]	EN_NONINTERLACE	Input	To choice the mode of interlace or non-interlace 0 :: interlace 1 :: non-interlace	0b
[2:0]	MODE	Input	For setting up the ENCODE mode 000 :: NTSC M 001 :: NTSC J 010 :: NTSC 4.43 011 :: M PAL 100 :: PAL(B, D, G, H and I) 101 :: PAL Nc 110 :: PAL N	000b

1.4.3. Internal control input

Table 1-5 Internal control input Description

Bits	Name	R/W	Description	Init. Value
[1:0]	LUMA_FILTER_SEL	input	Select of the luminance filter 00 :: 6MHz 01 :: NTSC notch 10 :: PAL notch 11 :: By pass	00b
[1:0]	CHRO_FILTER_SEL	Input	Select of the chrominance filter 00 :: 1.35 Mhz 10 :: 0.67 Mhz	00b
[0]	EN_COLOR_KILL	Input	Color information is not output to composite DAC (only luminance output) 0 :: disable 1 :: enable	0b
[0]	EN_REST_SCH	Input	This bit can be auto reset SCH 0 :: disable 1 :: Enable auto reset Resetting the SCH phase every four or eight fields. This function is to avoid the accumulation of SCH phase error	0b
[0]	EN_INTERNAL_PATTERN	Input	Internal pattern generation setting 0 :: disable 1 :: pattern generation	0b

Bits	Name	R/W	Description	Init. Value
[2:0]	COLOR_PATTERN_MODE	Input	000 :: Colorbar 001 :: Red lamp 010 :: Green lamp 011 :: Blue lamp	000b
[2:0]	CHRO_DELAY	Input	For setting the delay of chrominance 000 :: none 001 :: 1 Ts delay 010 :: 2 Ts delay 110 :: 6 Ts delay 111 :: 7 Ts delay (1Ts – 1clock period :: 27Mhz → about 37ns)	000b
[2:0]	LUMA_DELAY	Input	For setting the delay of luminance 000 :: none 001 :: 1 Ts delay 010 :: 2 Ts delay 110 :: 6 Ts delay 111 :: 7 Ts delay (1Ts – 1clock period :: 27Mhz → about 37ns)	000b
[1:0]	BURST_WID	Input	For setting burst falling to rising and rising to falling edge time 00 :: 1 Ts reduce from normal 01 :: normal 10 :: 1 Ts expanded	01b
[2:0]	HSYNC_WID	Input	For setting the delay of luminance 000 :: 2 Ts reduce from normal mode 001 :: 1 Ts reduce from normal mode 010 :: normal mode 011 :: 1 Ts expanded 100 :: 2 Ts expanded (1Ts – 1clock period :: 27Mhz → about 37ns)	010b

1.4.4. Subcarrier phase input

Table 1-6 subcarrier phase Register

Bits	Name	R/W	Description	Init. Value
[15:0]	SUB_PHASE	R/W	Subcarrier phase register Each bit represents 0.0055°	0000h

1.4.5. Subcarrier input

Table 1-7 subcarrier input

Bits	Name	I/O	Description	Init. Value
[31:0]	SUB_REQ	Input	Subcarrier frequency setting	0000h

1.4.6. Image Control register

Table 1-8 Image control Register

Bits	Name	R/W	Description	Init. Value
[7:0]	SATURATION_YLEV	R/W	Luminance Saturation level control register. If you not use saturation effect, you setting 80h(128) NTSC recommend value :: 144 PAL recommend value :: 136	00h
[7:0]	SATURATION_CLEV	R/W	Chrominance Saturation level control register. If you not use saturation effect, you setting 80h(128) NTSC recommend value :: 144 PAL recommend value :: 136	00h
[7:0]	HUE_LEV	R/W	Hue level control(0 ~ 255). Each bit has 1.41 degree value Setting 0, which is 0° shift Setting 1, which is 1.41° shift	00h
[4:0]	BRIGHTNESS_LEV	R/W	Brightness level control(-32 ~ 31)	00h

1.4.7. Output level register 0

Table 17-9 Output level Register 0 (Offset : 0018h)

Bits	Name	R/W	Description	Init. Value
[9:0]	BLACK_LEV	R/W	BLACK level control register. Recommend setting value :: NTSC → 282 / PAL → 252	00h
[9:0]	BLANK_LEV	R/W	BLANK Level control register. Recommend setting value : NTSC → 240 / PAL → 252	00h

1.4.8. Output level register 1

Table 17-10 Output level Register 1 (Offset : 001Ch)

Bits	Name	R/W	Description	Init. Value
[7:0]	HSYNC_STEP	R/W	2's complement value NTSC :: 8'hC8 PAL :: 8'hDE	0b
[7:0]	BURST_CAL	R/W	This bit is for calibration of Burst level. This level range 0 to 1.99x. the setting level 128 is the range value 1	10000000b
[4:0]	BURST_STEP	R/W	For burst level setting, NTSC :: 14 PAL :: 15	0b